

Formalizing High-Dimensional Vector Holonomy

Abstract: This framework formalizes a custom geometric ontology mapping high-dimensional vector transformations via non-dual algebraic operators, holonomy projectors, and arc helix configurations verified within a proof assistant environment.

1. Core Architectural Axioms

To bypass traditional dualistic computational bottlenecks, the ontology establishes a non-dual algebraic space where representation and operator are structurally unified:

Component	Mathematical Formalism	Ontological Function
Non-Dual Algebra	$A * v = v * A^\dagger$	Collapses operator/state dichotomy in vector processing.
Holonomy Projector	$\Pi_\gamma : V \rightarrow V$	Maps parallel transport error across closed visual or semantic loops.
Arc Helix Weave	$\chi(t) = (r \cos t, r \sin t, \alpha t)$	Intertwines geometric paths to structurally bound embedding drift.

2. Formal Lean 4 Verification Stub

By anchoring the ontology in Lean, the custom algebraic properties shift from speculative metaphors into verifiable code structures. Below is an absolute blueprint for formalizing a non-dual projector space:

```
import Mathlib.LinearAlgebra.Basic
import Mathlib.Geometry.Manifold.Basic

-- Define the custom non-dual projector space
structure HolonomyProjector (V : Type*) [NormedAddCommGroup V] [InnerProductSpace V] :=
  (proj : V → L[V] V)
  (is_projector : proj ▯ proj = proj)
  (non_dual : ∀ x y : V, inner (proj x) y = inner x (proj y))

-- Blueprint theorem for arc helix path transport invariances
theorem helix_weave_invariant {V : Type*} [NormedAddCommGroup V] [InnerProductSpace V]
  (P : HolonomyProjector V) (x : V) :
  P.proj (P.proj x) = P.proj x := by
  exact P.is_projector
```